PRD : Fact Checker & AI Image Analyzer

Project Description:

- In the current digital age and with the rising use of Artificial Intelligence (AI), misinformation is very common. It can be difficult and time consuming to authenticate claims and images posted online. Our app serves the role of an AI Fact-Checker and Image Analyzer for users to regain clarity in today's media. The aim is to fact-check given claims by the user and users can upload two key forms of media: png/jpg (image) upload, or web article URL. For added depth we also wanted to implement a function to determine if images are AI altered or generated.

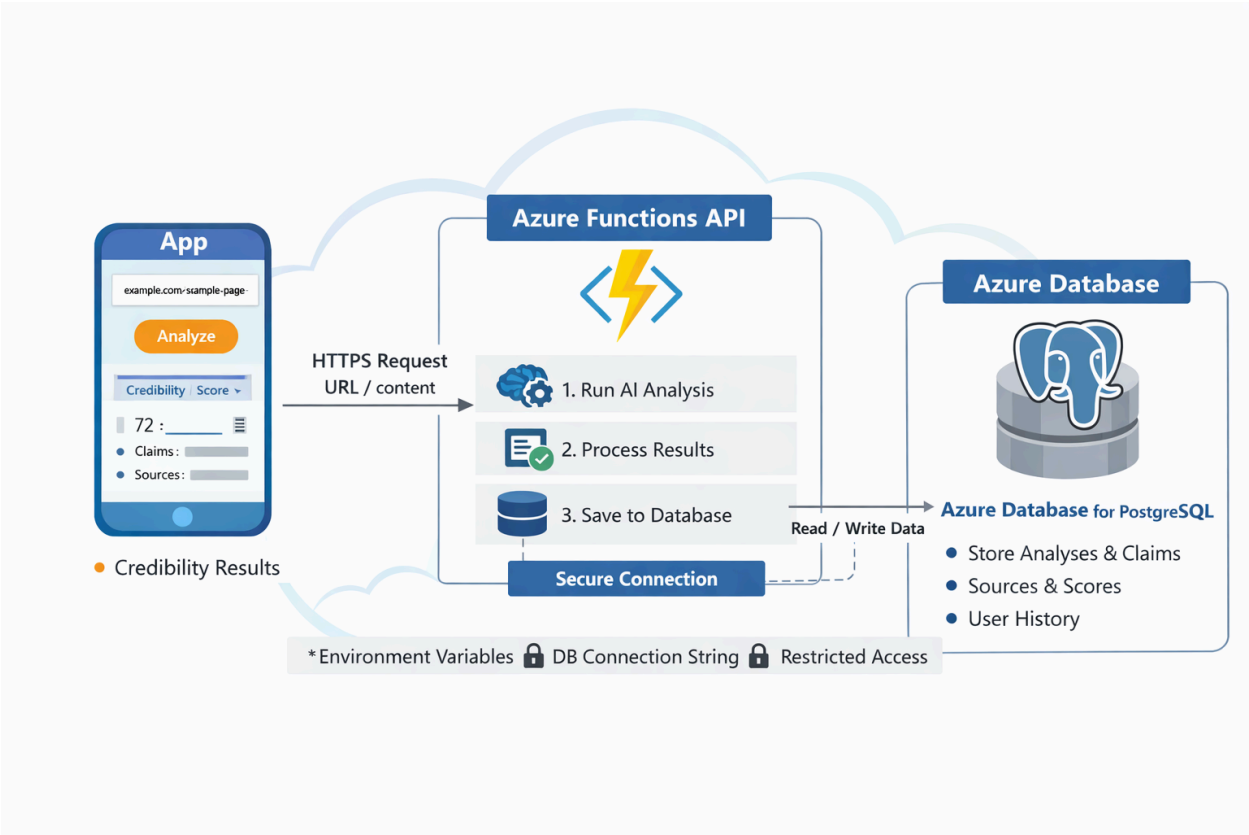
Required Features:

- The user must be able to upload a link to a webpage for AI analysis. The link should be to a source that is on the internet
- The user must be able to upload an image for AI analysis. The screenshot should be in standardized format of .png/.jpg
- The user must be able to talk to a generalized AI chatbot specialized in fact-checking user's questions with credible claims and sources.
- The AI will ingest the given link and determine a credibility score of the given claims and statements in the linked webpage/article. The credibility analysis will be proven with factual links found by the AI and reasoned to the user.
- General AI Agent Type(Agentic AI)

Aspirational Features:

- Targeted focus on Instagram, or other social media, where fake news and misinformation are generally more prevalent.
- Allow the user to upload videos and images to see if they are AI generated.

Architecture Overview:

- The system follows a cloud-based client–server architecture hosted entirely on Microsoft Azure.
- The client application (App) allows users to submit a webpage URL for fake news analysis and displays the credibility results.
- The app communicates with the backend through HTTPS requests and does not perform any AI processing or database access locally.
- The system shall use Azure Database for PostgreSQL as a managed cloud database to store all analysis data.
- The database shall store structured information including analyzed URLs, credibility scores, AI summaries, extracted claims, sources, and user history.
- The database shall be accessed only by the Azure Functions backend through secure connection strings stored as environment variables; the client app shall not access the database directly.

- The backend shall receive webpage URLs from the client app via an Azure Functions API, extract the webpage content, and perform AI-based credibility analysis.
- The backend shall process and store the analysis results in Azure Database for PostgreSQL and return the credibility score and report to the client app.